Mthunzelwa Makwanyane
20241063
NET731 Practicals – Week2

CTU Training Solutions

NET731 – Networking 3A

Week 2 Practical: Cloud Network Design

Reasons for configuration choices

The cloud provider I used for this activity is Azure because it is a very trustworthy and efficient identity provider used by plenty of organizations world-wide. Since the business needs an internal network for the virtual network, I used the CIDR address range: 10.0.0.0 which includes 65,536 addresses in total. The first subnet address is 10.0.1.0, containing 256 addresses and I named it Public addresses subnet. I gave this subnet that specific number of addresses considering any expansion of other resources such as virtual machines. The second subnet address is 10.0.2.0/26, this means there will only be 64 assignable addresses. I did that due to the organization size being small and the fact that backend servers do not need a significant amount of address spaces.

The two NSGs I created are called TechBridgepublic and TechBridgeprivate, one for public internet access and one for private traffic only for the backend subnet. For the TechBridgepublic NSG, I configured it for public traffic to enter the VNet by setting the destination port ranges to 80 and 443, setting the protocol to TCP and clicking the allow traffic option. I also set the source to any with the asterisk sign to allow internet (public) traffic. For the TechBridgeprivate NSG, I configured the source to a service tag and selecting the Virtualnetwork option. Using the Virtual network as a source makes sure that the address spaces within the VNet can communicate with each other. For both of these NSGs, I configured their inbound settings. I implemented the TechBridgepublic to the Public address subnet and the TechBridgeprivate to the Backend subnet.

Creating Virtual network and Subnets

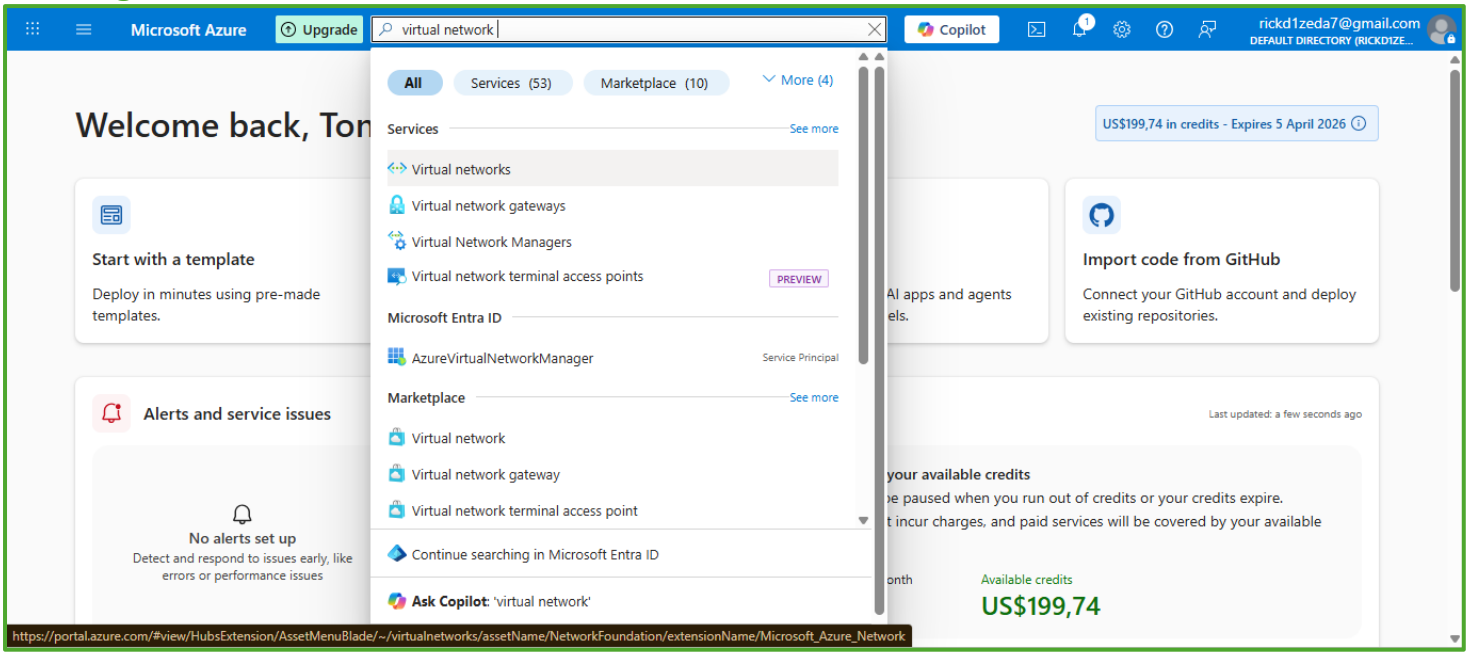

Figure 1: Searching for virtual network resource

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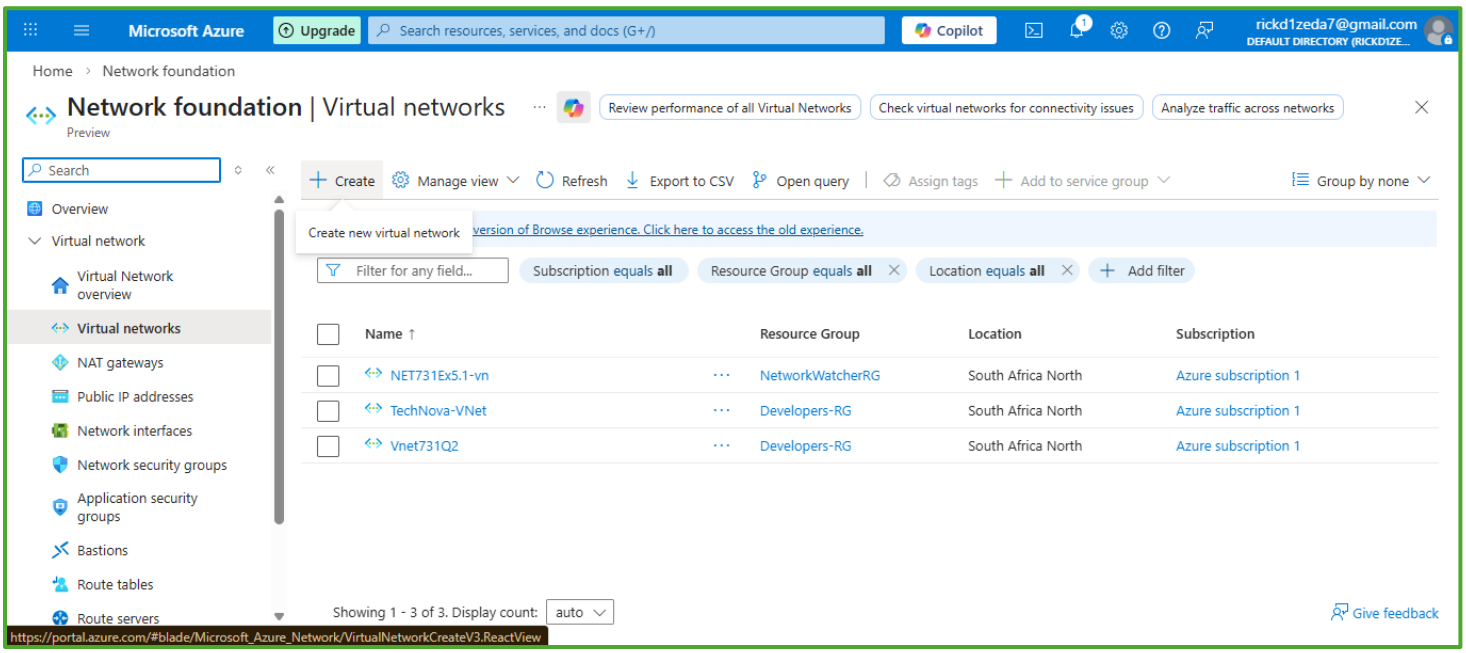

Figure 2: Creating a new virtual network

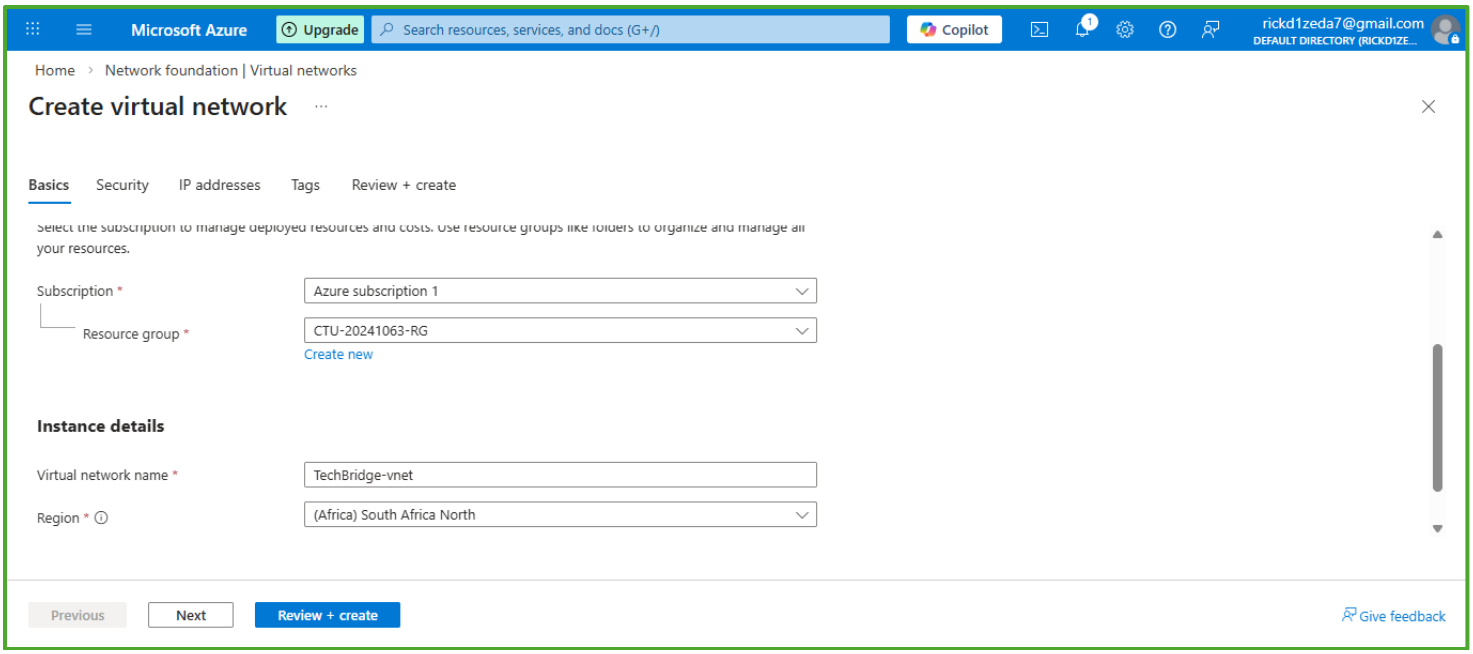

Figure 3: Basic virtual network information

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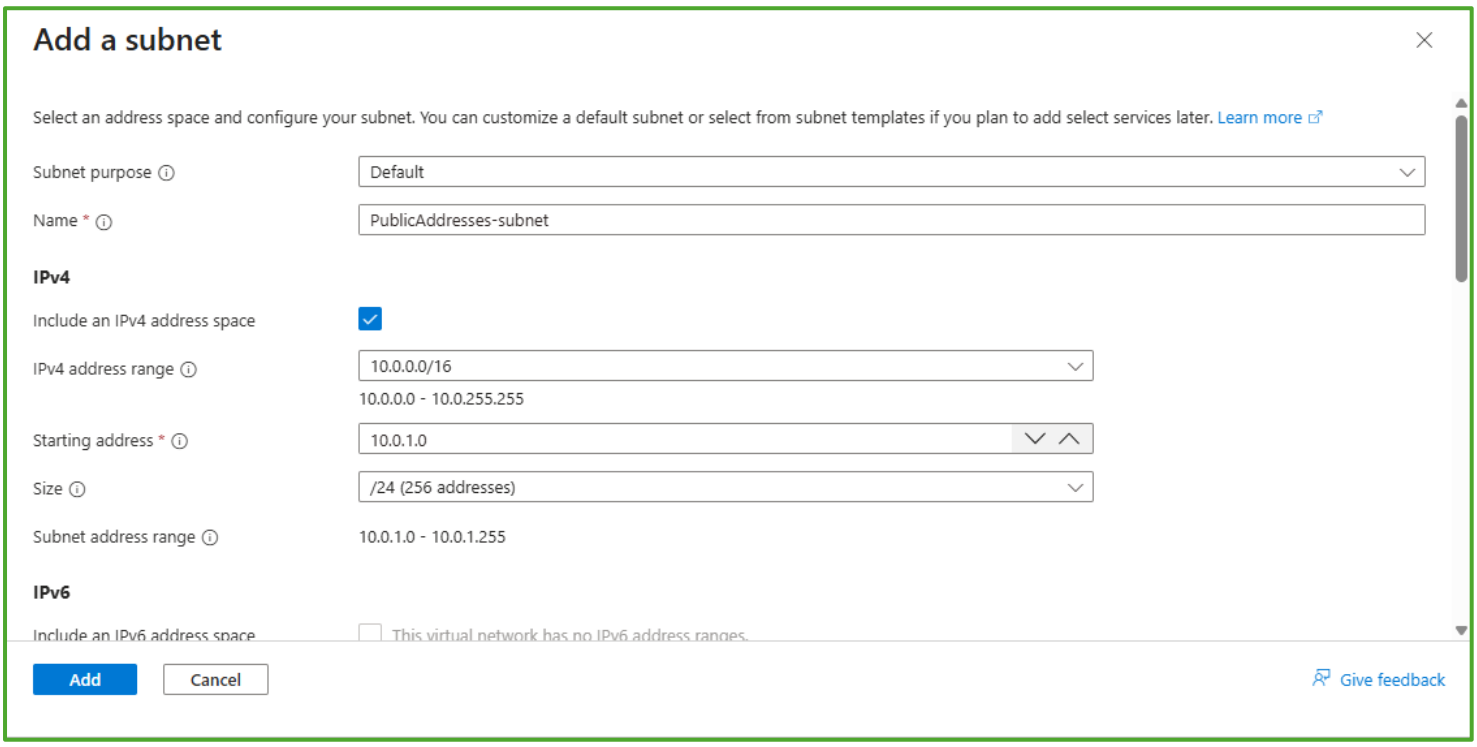

Figure 4: First subnet configuration

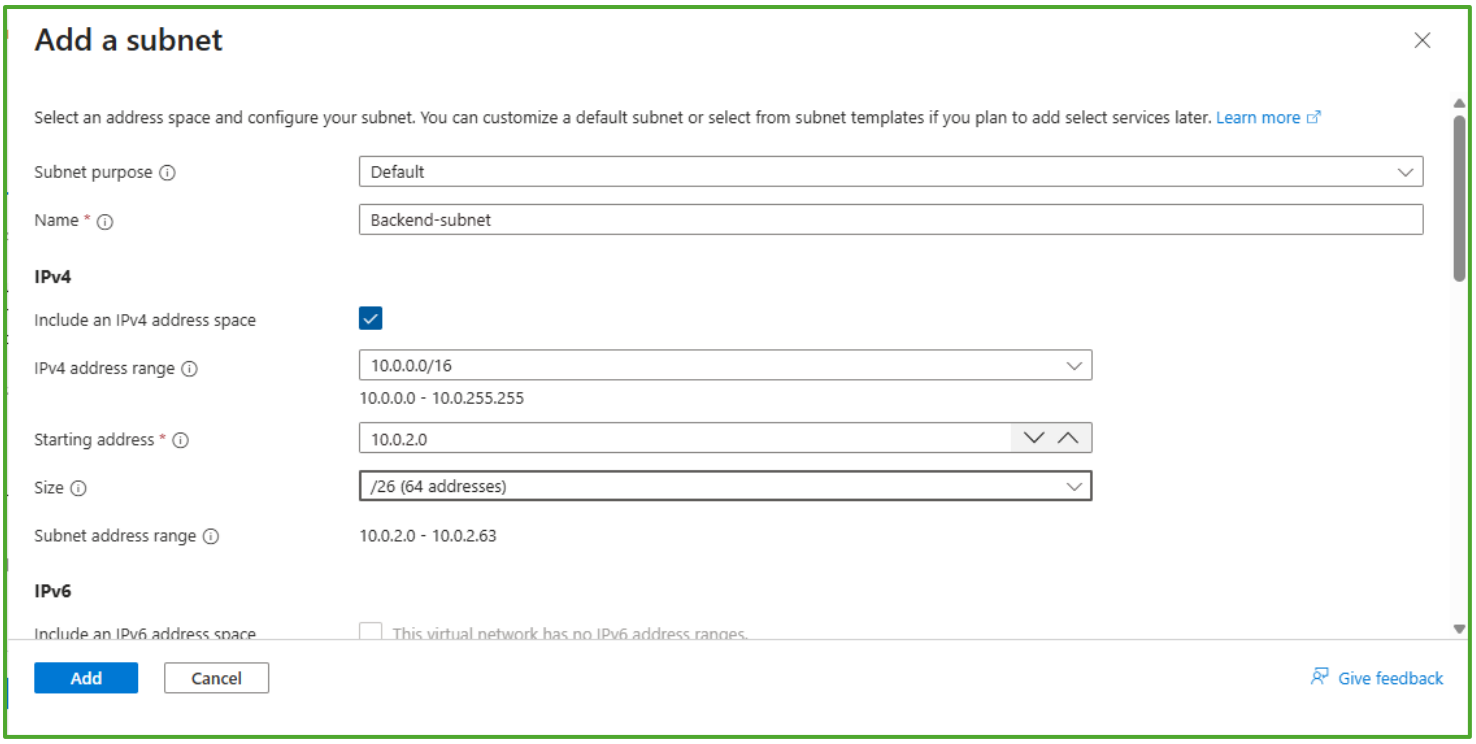

Figure 5: Second subnet configuration

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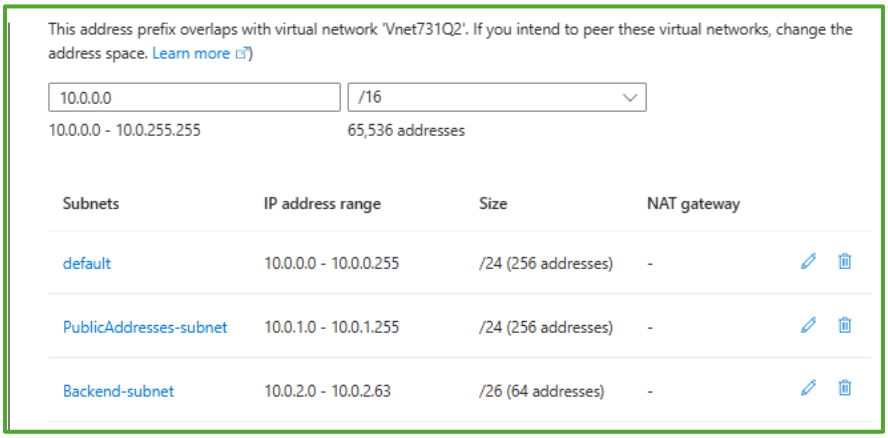

Figure 6: Ip address and subnets

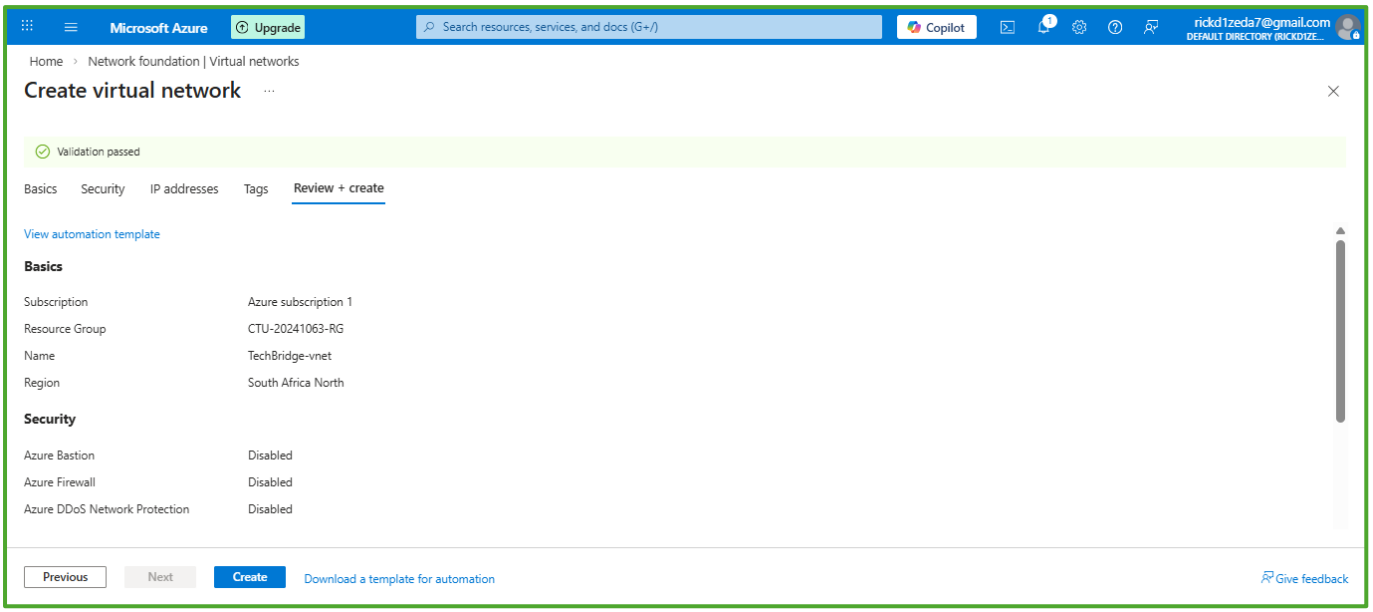

Figure 7: Reviewing and creating the virtual machine

Creating the NSGs

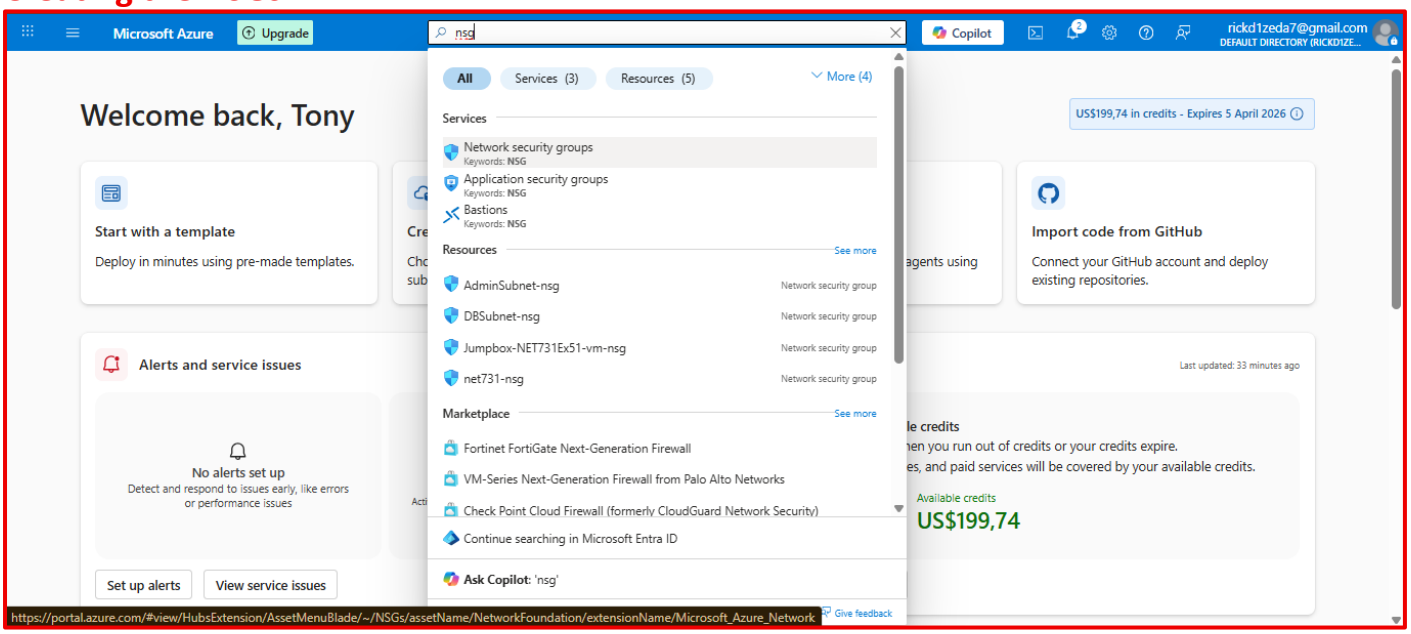

Figure 8: Searching for NSG resource

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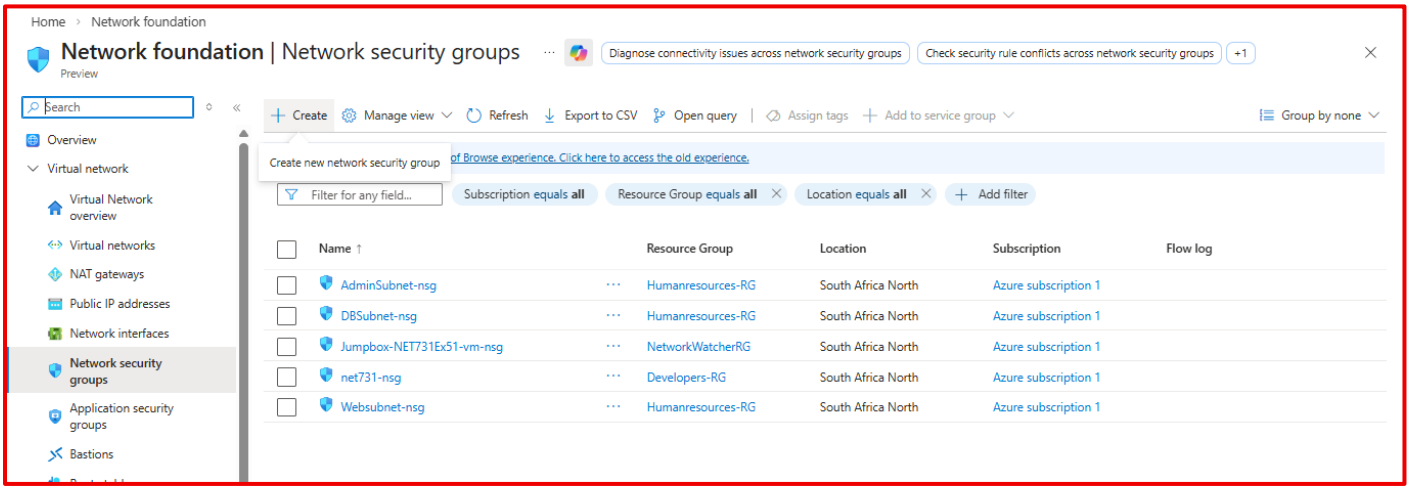

Figure 9: Creating a new NSG

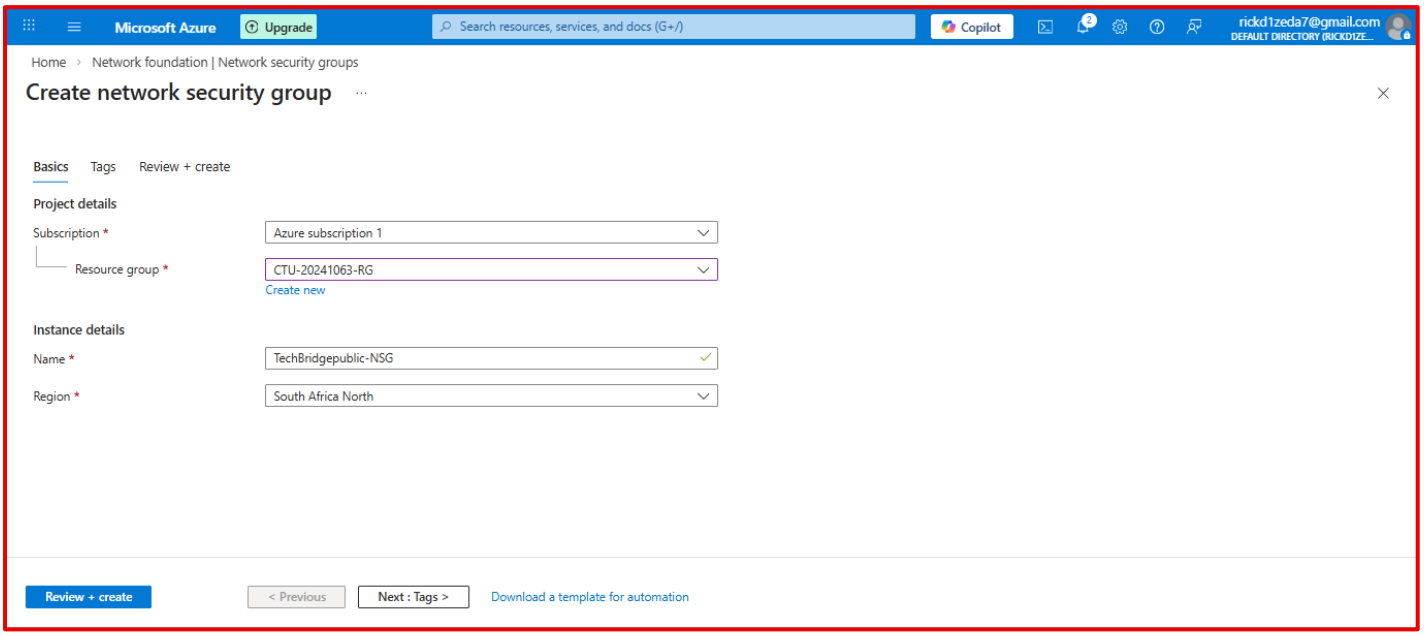

Figure 10: Creation of first NSG

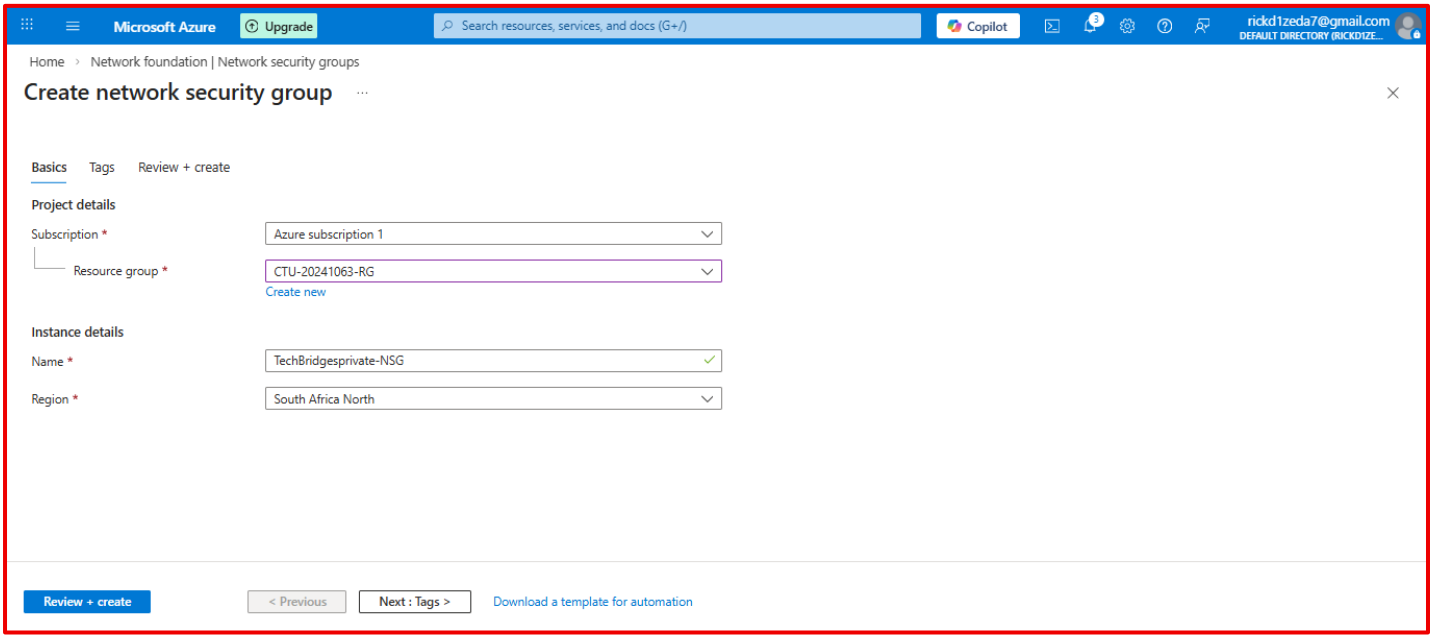

Figure 11: Creation of second NSG

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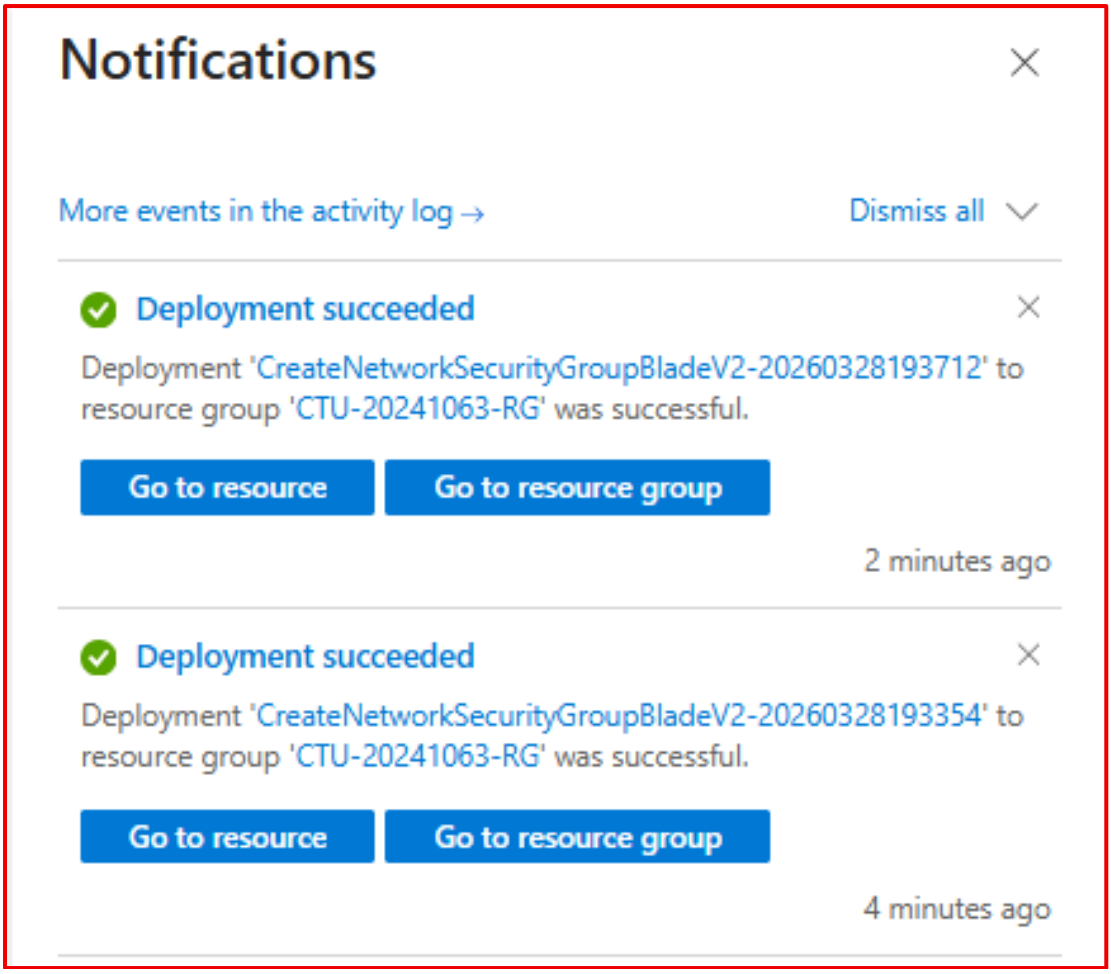

Figure 12: Successful deployment of both NSGs

Configuring NSGs

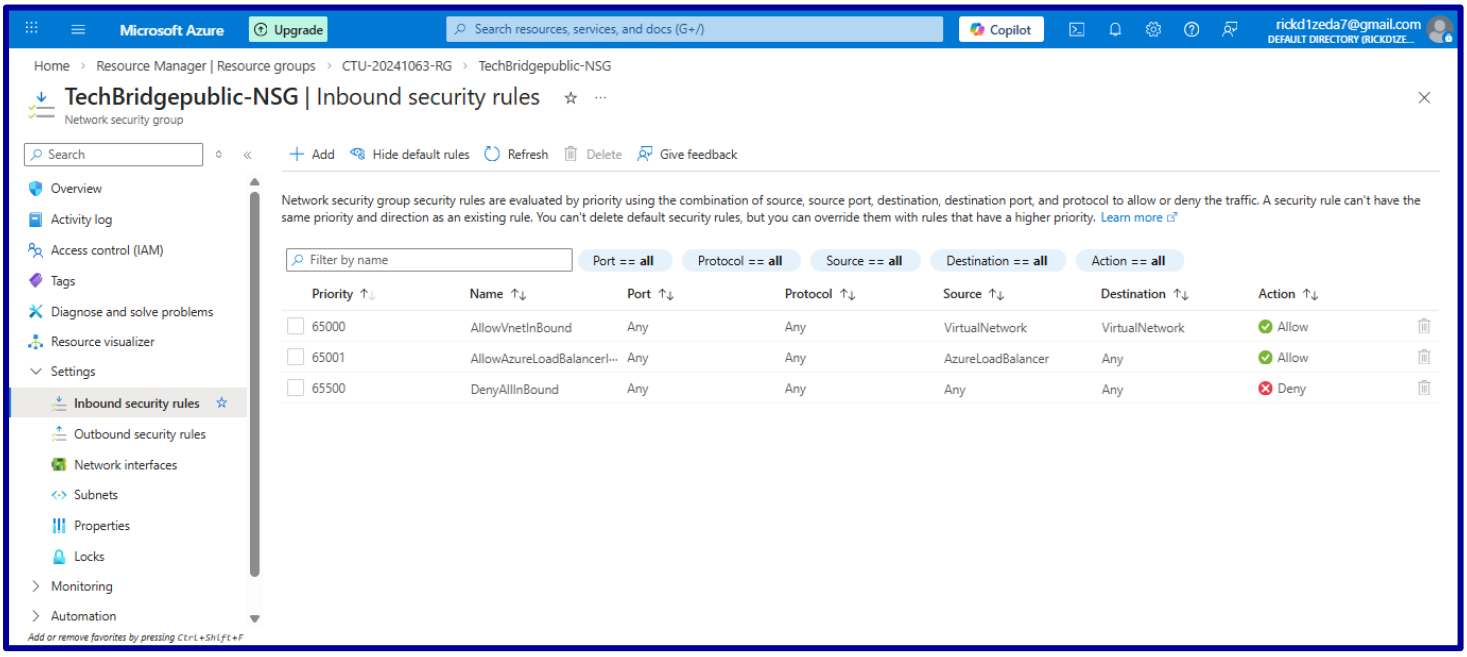

Figure 13: Adding new inbound rule in TechBridgepublic NSG

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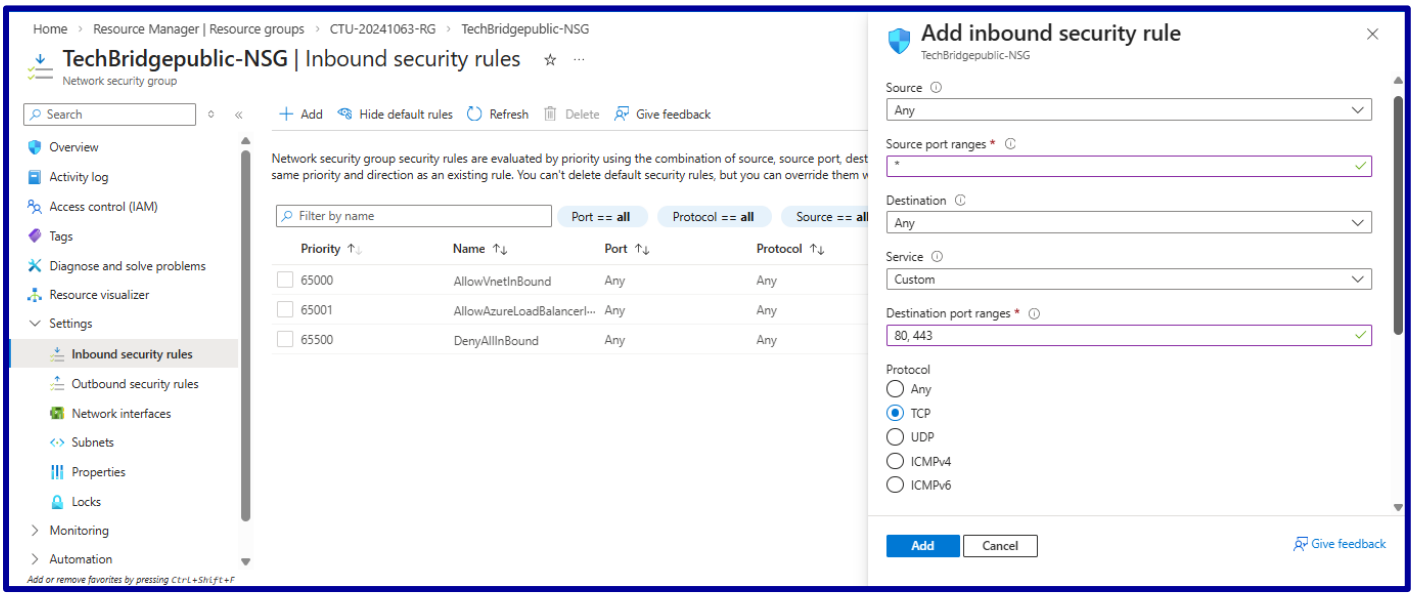

Figure 6: Configuring new inbound settings

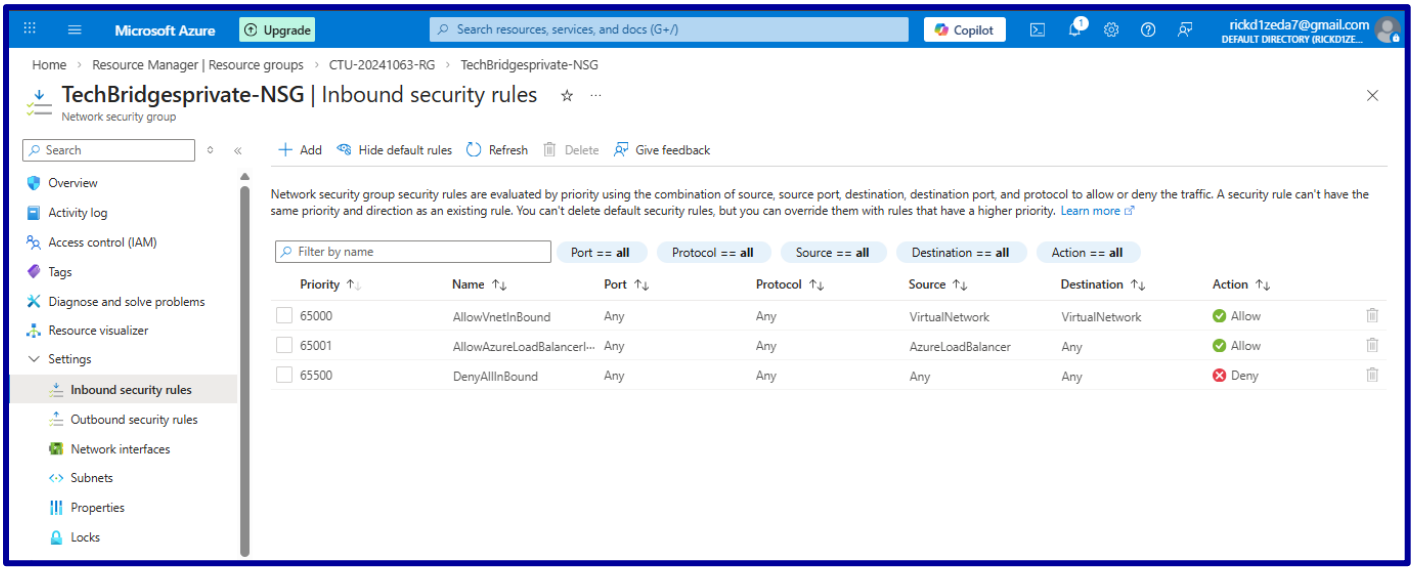

Figure 7: Adding new inbound security rule in TechBridgeprivate NSG

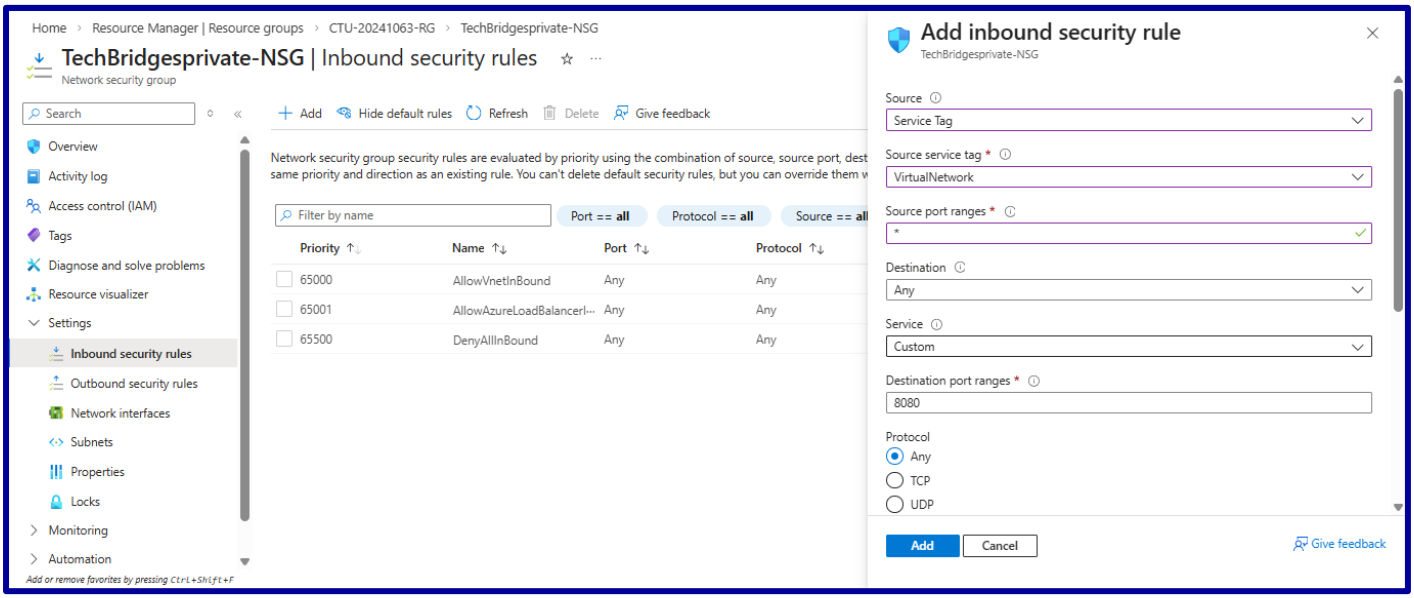

Figure 8: Configuring the new NSG rules

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Applying the NSGs to the subnets

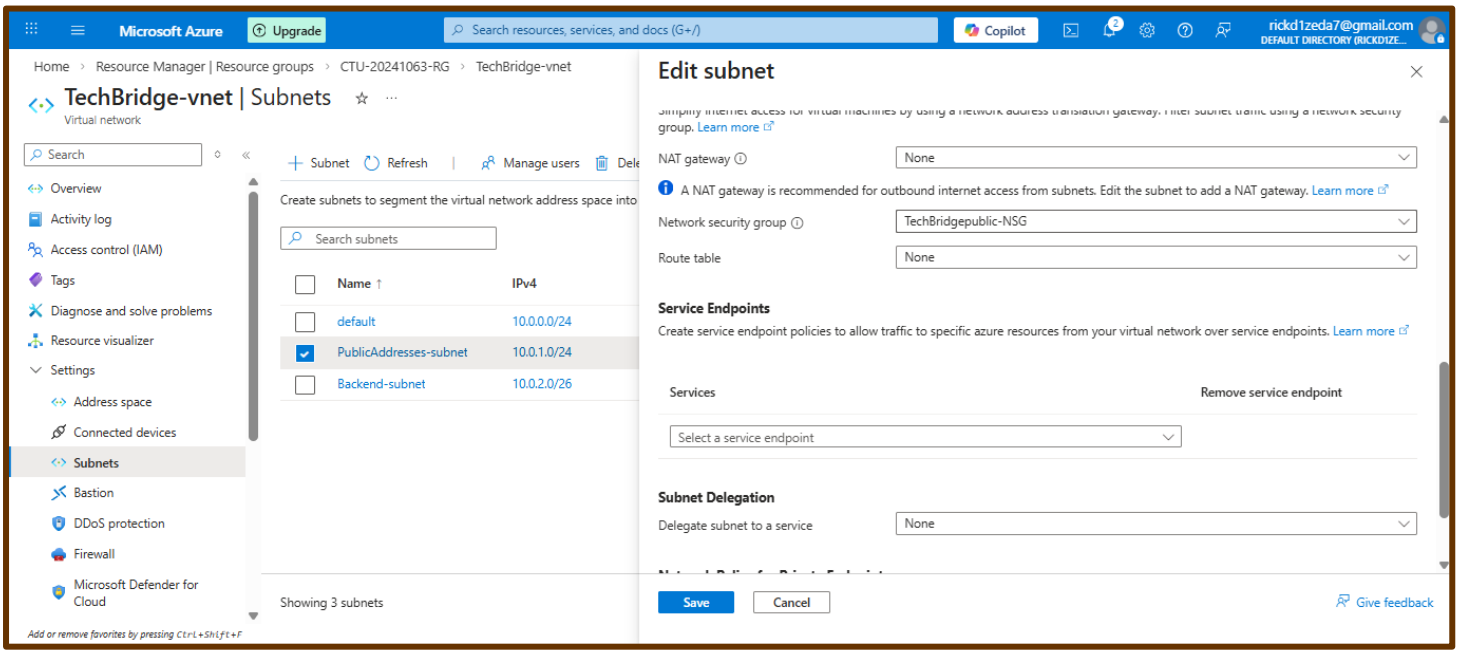

Figure 9: Applying TechBridgepublic NSG to PublicAddresses subnet

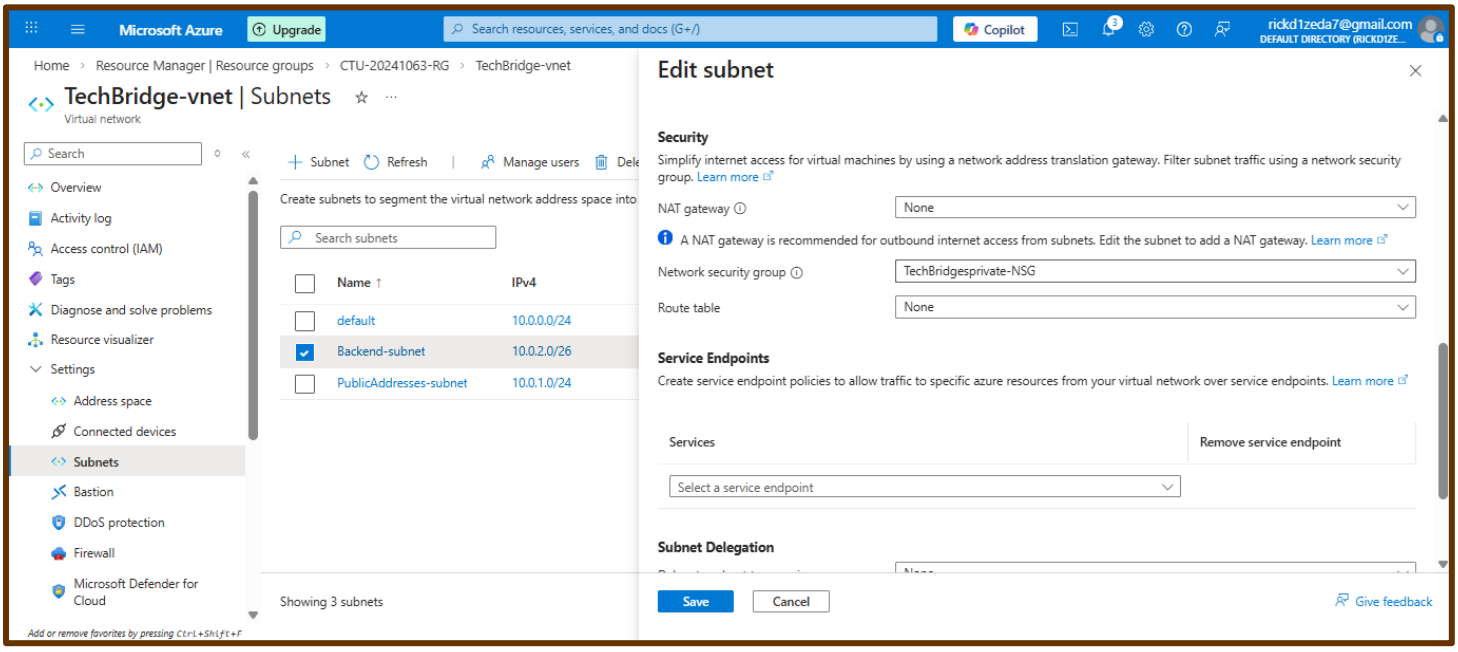

Figure 10: Applying the TechBridgeprivate NSG to the Backend subnet

Task 3 – Identify Required Azure Services

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Azure Service	Purpose / Why you need it	Week it is used
Resource Group	For better management of resources	Week 1
Virtual Network (VNet)	For secure communication of azure resources like virtual machines	Week 2
Virtual Machine (VM – B1s)		
Microsoft Entra ID		
Network Security Group (NSG)		
Azure Storage Account		
Microsoft Defender for Cloud		
Azure Monitor		
Azure Cost Management		